

[별표 1] 방사성핵종별 자체처분 허용농도(제2조제2호 관련)

방사성핵종	허용농도 (Bq/g)
I-129	0.01
Na-22, Sc-46, Mn-54, Co-56, Co-60, Zn-65, Nb-94, Ru-106, Ag-110m, Sb-125, Cs-134, Cs-137, Eu-152, Eu-154, Ta-182, Bi-207, Th-229, U-232, Pu-238, Pu-239, Pu-240, Pu-242, Pu-244, Am-241, Am-242m, Am-243, Cm-245, Cm-246, Cm-247, Cm-248, Cf-249, Cf-251, Es-254	0.1
C-14, Na-24, Cl-36, Sc-48, V-48, Mn-52, Fe-59, Co-57, Co-58, Se-75, Br-82, Sr-85, Sr-90, Zr-95, Nb-95, Tc-96, Tc-99, Ru-103, Ag-105, Cd-109, Sn-113, Sb-124, Te-123m, Te-132, Cs-136, Ba-140, La-140, Ce-139, Eu-155, Tb-160, Hf-181, Os-185, Ir-190, Ir-192, Tl-204, Bi-206, U-233, U-234, U-235, U-238, Np-237, Pu-236, Cm-243, Cm-244, Cf-248, Cf-250, Cf-252, Cf-254	1
Be-7, F-18, Cl-38, K-43, Ca-47, Mn-51, Mn-52m, Mn-56, Fe-52, Co-55, Co-62m, Ni-65, Zn-69m, Ga-72, As-74, As-76, Sr-91, Sr-92, Zr-93, Zr-97, Nb-93m, Nb-97, Nb-98, Mo-90, Mo-93, Mo-99, Mo-101, Tc-97, Ru-97, Ru-105, Cd-115, In-111, In-114m, Sn-125, Sb-122, Te-127m, Te-129m, Te-131m, Te-133, Te-133m, Te-134, I-126, I-130, I-131, I-132, I-133, I-134, I-135, Cs-129, Cs-132, Cs-138, Ba-131, Ce-143, Ce-144, Gd-153, W-181, W-187, Pt-191, Au-198, Hg-203, Tl-200, Tl-202, Pb-203, Po-203, Po-205, Po-207, Ra-225, Pa-230, Pa-233, U-230b, U-236, Np-240, Pu-241, Cm-242, Es-254m	10
H-3, S-35, K-42, Ca-45, Sc-47, Cr-51, Mn-53, Co-61, Ni-59, Ni-63, Cu-64, Rb-86, Sr-85m, Sr-87m, Y-91, Y-91m, Y-92, Y-93, Tc-97m, Tc-99m, Rh-105, Pd-109, Ag-111, Cd-115m, In-113m, In-115m, Te-129, Te-131, I-123, I-125, Cs-135, Ce-141, Pr-142, Nd-147, Nd-149, Sm-153, Eu-152m, Gd-159, Dy-166, Ho-166, Er-171, Tm-170, Yb-175, Lu-177, Re-188, Os-191, Os-193, Ir-194, Pt-197m, Au-199, Hg-197, Hg-197m, Tl-201, Ra-227, U-231, U-237, U-239, U-240, Np-239, Pu-234, Pu-235, Pu-237, Bk-249, Cf-253, Es-253, Fm-255	100
Si-31, P-32, P-33, Fe-55, Co-60m, Zn-69, As-73, As-77, Sr-89, Y-90, Tc-96m, Pd-103, Te-125m, Te-127, Cs-131, Cs-134m, Pr-143, Pm-147, Pm-149, Sm-151, Dy-165, Er-169, Tm-171, W-185, Re-186, Os-191m, Pt-193m, Pt-197, At-211, Th-226, Pu-243, Am-242, Cf-246	1,000
Co-58m, Ge-71, Rh-103m, Fm-254	10,000

주1) 다수의 방사성핵종이 혼합되어 있는 경우에는 다음과 같이 한다.

$$\sum_i \frac{C_i}{C_{L,i}} < 1$$

C_i : 방사성핵종 i 의 방사능농도(Bq/g)

$C_{L,i}$: 별표1에 주어진 방사성핵종 i 의 자체처분 허용농도(Bq/g)

주2) 별표1에 수록되지 않은 알파선을 방출하지 아니하는 방사성핵종의 경우 자체처분 허용농도로서 0.1 Bq/g을 적용할 수 있다.

주3) 다음 목록에 제시된 모핵종과 그 모핵종의 붕괴로 생성되는 자핵종이 함께 존재하는 경우, 모핵종에 대해서만 허용농도를 적용한다.

모핵종	자핵종
Fe-52	Mn-52m
Zn-69m	Zn-69

Sr-90	Y-90
Sr-91	Y-91m
Zr-95	Nb-95
Zr-97	Nb-97m, Nb-97
Nb-97	Nb-97m
Mo-99	Tc-99m
Mo-101	Tc-101
Ru-103	Rh-103m
Ru-105	Rh-105m
Ru-106	Rh-106
Pd-103	Rh-103m
Pd-109	Ag-109m
Ag-110m	Ag-110
Cd-109	Ag-109m
Cd-115	In-115m
Cd-115m	In-115m
In-114m	In-114
Sn-113	In-113m
Sb-125	Te-125m
Te-127m	Te-127
Te-129m	Te-129
Te-131m	Te-131
Te-132	I-132
Cs-137	Ba-137m
Ce-144	Pr-144, Pr-144m
U-232	Th-228, Ra-224, Rn-220, Po-216, Pb-212, Bi-212, Tl-208
U-240	Np-240m, Np-240
Np-237	Pa-233
Pu-244	U-240, Np-240m, Np-240
Am-242m	Np-238
Am-243	Np-239
Cm-247	Pu-243
Es-254	Bk-250
Es-254m	Fm-254